

CS214-Data Structure

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That course is inspired from the following Drs:

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Good Software

Essential attributes of good software



Product characteristic	Description
Maintainability	Software should be written in such a way so that it can evolve to meet the changing needs of customers. This is a critical attribute because software change is an inevitable requirement of a changing business environment.
Dependability and security	Software dependability includes a range of characteristics including reliability, security and safety. Dependable software should not cause physical or economic damage in the event of system failure. Malicious users should not be able to access or damage the system.
Efficiency	Software should not make wasteful use of system resources such as memory and processor cycles. Efficiency therefore includes responsiveness, processing time, memory utilisation, etc.
Acceptability	Software must be acceptable to the type of users for which it is designed. This means that it must be understandable, usable and compatible with other systems

Skills of Software engineer

Skills You need to learn as a Software Engineer



ABILITY TO WRITE, READ AND
FOLLOW DIRECTIONS



ABILITY TO UNDERSTAND AND
FOLLOW DESIGN PATTERNS



ABILITY TO WRITE EFFICIENT
ALGORITHMS



ABILITY TO WORK WITH OR
MANAGE A TEAM

Efficiency of an Algorithm

- An algorithm which **takes fewer resources** and computes results in a **minimum time** for a problem then that algorithm is known as efficient.

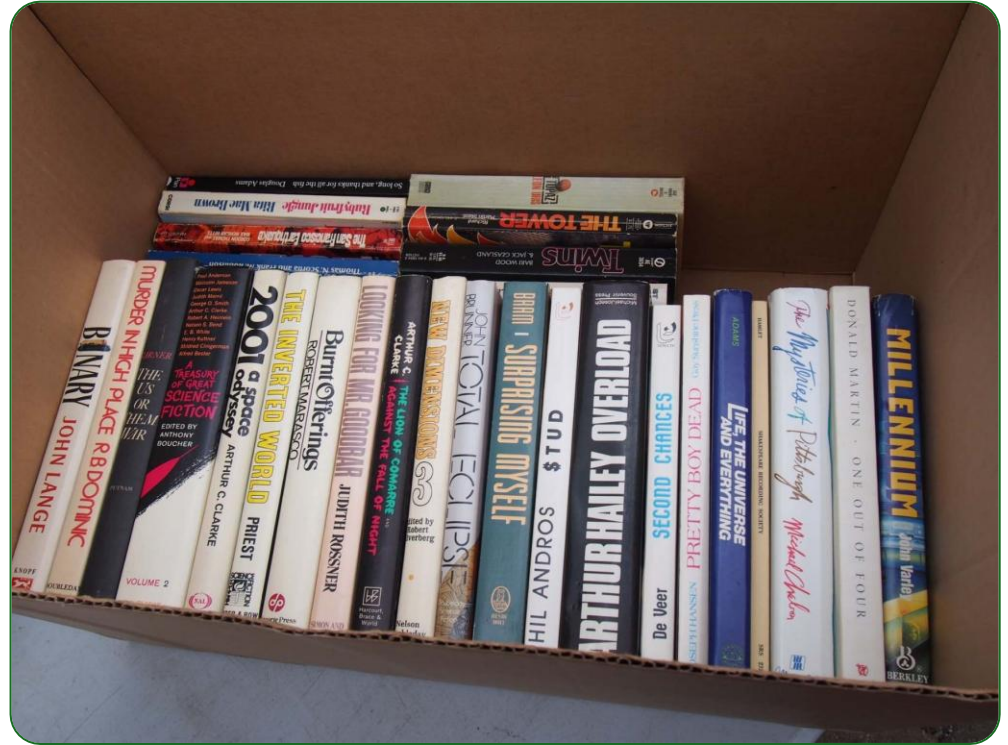
Data UnStructure



Data Structure



Data Structure



Data Structure

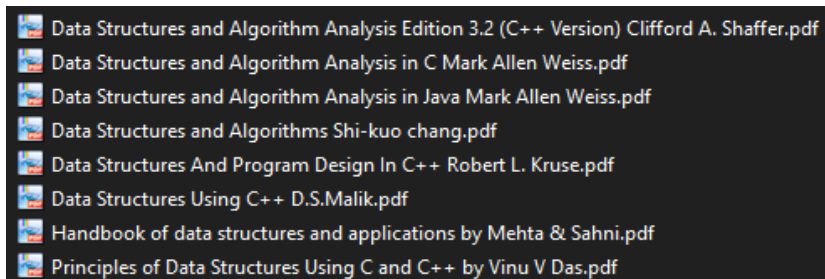


Data Structure



Textbook

- Very Important Texts in C
 - Kernighan and Ritchie (1988) is a very good text, but not for beginners.
- Very Important Texts in Programming (Algorithms and Data Structures)
 - Knuth (1997) is a reference by Knuth, the father of computer programming (the first 3 volumes are dedicated to Algorithms, along with their analyses, and Data structures). Every student in computer science has to read from this book. Sedgewick (1998) is a reference by Sedgewick, a student of Knuth. It is a very good text, full of insight, in algorithms and data structures in C. The author has the book in different programming languages, e.g., C++, and Java. Cormen (2001) is a key book in algorithms in pseudo-code. Press (1992) is a great source in numerical recipes in C, with ready-to-use code.



Syllabus

Week1	Introduction	
Week2	Stack	Sheet1: Stack
Week3	Queue	Sheet2: Queue Sheet3: Stack & Queue
Week4	Linked list	Sheet4: Linked list
Week5	Linked stack and queue	Sheet #5 Linked Stack & Queue Quiz1
Week6	Linked list use cases	Sheet #6 Linked Lists Use Cases
Week7	Midterm	
Week8	Trees	Sheet #7 Trees
Week9	Binary Search Tree	Sheet #8 Binary Search Trees
Week10	AVL Tree	Sheet #9 AVL Trees Quiz2
Week11	Hashing	Sheet #10 Hashing
Week12	Revision	



Grades Distribution

- Grades Distribution:
 - Final Written Exam **50%**
 - Midterm Written Exam **25%**
 - Sheets **10%**
 - Quizzes **5%**
 - Practical **10%**

